CampusAI EnergyWise

AI-Powered Campus Energy Optimization & Sustainability Advisor

1M1B AI for Sustainability Virtual Internship

Submitted by: Shwetanjali Gautam

Institution: Birla Institute of Technology, Mesra

Primary SDG: SDG 7 — Affordable and Clean Energy

1. Executive Summary

CampusAI EnergyWise is an AI-enabled decision-support concept for improving energy efficiency in educational campuses. It combines structured energy observations with occupancy and temperature context, applies predictive/anomaly logic, and uses generative AI to translate observations into understandable sustainability recommendations. The system is human-in-the-loop: AI identifies patterns and proposes actions, while facility personnel approve operational interventions.

2. Problem Definition

Campus buildings consume electricity for lighting, cooling, laboratories, computing and residential services. Raw energy measurements can reveal consumption levels but do not automatically explain why a value is unusual or what a responsible next step should be. This project bridges that gap with a lightweight AI workflow.

3. SDG Alignment

The primary alignment is UN SDG 7, Affordable and Clean Energy, particularly Target 7.3 on energy efficiency. The project also supports SDG 12 through efficient resource consumption. SDG 13 is supporting because reduced electricity demand can contribute to lower emissions when a documented electricity emissions factor is applied. No specific carbon reduction is claimed without measured data.

4. Design Thinking

Empathize: facility teams need understandable information rather than another raw data feed. Define: the gap is between energy observations and timely, explainable action. Ideate: combine prediction, anomaly detection, generative explanation and automation. Prototype: use CSV observations, a Python model and an n8n workflow blueprint. Test: validate numerical performance and recommendation quality with human reviewers.

5. System Architecture

The pipeline is data source → input validation → feature preparation → predictive/anomaly model → AI explanation prompt → structured recommendation → human review → sustainability log/dashboard. Inputs include building, timestamp, occupancy, temperature and energy consumption.

6. AI Component

A Random Forest regression model is included for demonstration prediction using building identity, occupancy, temperature and day of week. A separate generative-AI stage explains observations and produces recommendations. This separation keeps numerical estimation and natural-language reasoning distinct.

7. Automation Workflow

The supplied n8n blueprint begins with a webhook, normalizes observations, constructs a responsible sustainability prompt, sends it to a configurable approved AI endpoint, and formats the report. The endpoint is intentionally configurable rather than embedding credentials or inventing API details.

8. Prototype Dataset

A 30-day synthetic dataset is supplied for Academic, Hostel and Laboratory Blocks. It contains daily occupancy, temperature and energy consumption. It is explicitly synthetic and must not be presented as actual BIT Mesra meter data.

9. Example AI Interaction

Example input: Laboratory Block, occupancy 22%, temperature 31.2 °C, energy 1,430 kWh/day. A responsible output could flag the case for review and suggest checking cooling schedules and laboratory equipment standby practices. It should not claim a specific saving without measurement.

10. Responsible AI

Fairness requires building-specific baselines. Transparency requires showing influential inputs. Privacy requires aggregate data rather than personal tracking. Safety requires AI to remain advisory. Data-quality safeguards should flag missing or implausible measurements. Human oversight is mandatory.

11. Evaluation Plan

The prototype can be evaluated with MAE and R^2 on held-out synthetic observations, but these are software demonstration metrics, not evidence of real-world savings. Real validation should use historical campus data, time-aware evaluation and measured post-intervention energy. Recommendations should be reviewed for relevance and safety.

12. Expected Impact

The intended impact is better interpretation of energy data, faster identification of review cases, and improved communication between technical teams and sustainability stakeholders. At scale, the system could support continuous monitoring and evidence-based scheduling or maintenance.

13. Limitations

The current project is a prototype using synthetic data. It is not connected to real campus meters and therefore does not establish actual energy or carbon savings. Weather, tariffs, equipment ratings, floor area and emissions factors are not included.

14. Future Scope

Future development can integrate smart meters, weather data, building-management systems, equipment metadata and dashboards. A retrieval-augmented layer could retrieve campus policies and equipment guidance before generating recommendations. A controlled pilot could measure energy intensity before and after approved interventions.

15. Conclusion

CampusAI EnergyWise demonstrates a practical AI-for-sustainability workflow that is understandable, scalable and responsible. It addresses a concrete campus problem, maps to SDG 7, uses AI for prediction and explanation, and preserves human authority over real-world decisions.

References

United Nations — Sustainable Development Goal 7: Affordable and Clean Energy — https://sdgs.un.org/goals/goal7

United Nations — Sustainable Development Goal 12: Responsible Consumption and Production — https://sdgs.un.org/goals/goal12

United Nations — Transforming our world: the 2030 Agenda for Sustainable Development — https://sdgs.un.org/2030agenda